Software Requirements Specification (SRS)

Project: Task Manager Application
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Revision History

Version	Date	Author	Change Summary	Approval
1.0	30-09-2026	Project Team	Initial SRS with use-case diagrams and RTM	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Project Guide			
Product Owner			

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Task Manager Application, a task management tool used to organise work and track its progress. It defines the functional and non-functional requirements, security requirements, external interfaces and verification criteria that developers, testers and evaluators will use as the baseline for design and testing.

1.2 Scope

The Task Manager Application is a web-based system in which individuals and small teams can create tasks, group them into projects, assign them to team members, set priorities and due dates, and monitor progress through a Kanban board and a dashboard. The scope includes user accounts, task and project management, reminders, activity history, collaboration through comments, CSV export and basic administration. It excludes native mobile apps, time-tracking and billing, third-party calendar integration and file attachments (see Future Work in the SAD).

1.3 Audience

Developers, QA engineers, system administrators, the product owner / project guide, and assessment evaluators.

1.4 Definitions

Term	Meaning
SRS	Software Requirements Specification
FR / NFR / SR	Functional / Non-Functional / Security Requirement
Task	A unit of work with title, description, due date, priority, status and assignee
Project	A container that groups tasks and members
Kanban board	Column view of tasks grouped by status
RBAC	Role-Based Access Control
JWT	JSON Web Token used for session authentication
REST / API	Representational State Transfer / Application Programming Interface
TLS	Transport Layer Security
WCAG	Web Content Accessibility Guidelines
RPO / RTO	Recovery Point / Recovery Time Objective
UI	User Interface

2. Overall description

2.1 Product perspective

The Task Manager is a standalone, browser-based application following a client-server model: a single-page web client communicates with a REST API server that stores data in a relational database. Email reminders and password-reset messages are sent through an external SMTP service. The system is self-contained and does not depend on other products.

2.2 Major product functions

- User registration, login, logout, password reset
- Create, edit, delete/restore tasks with priority, due date and status
- Projects with members, categories and tags

- Assign tasks to project members
- Search, filter and sort tasks
- Kanban board and progress dashboard
- Due-date reminders (in-app and email)
- Task comments and activity history
- CSV export of task lists
- Administration of users and roles

2.3 User roles and characteristics

- **Guest:** unauthenticated visitor; can only register, log in or reset a password.
- **Member (Registered User):** students, professionals or team members with basic computer skills who expect fast, uncluttered task entry and clear progress views.
- **Project Owner:** a Member who created a project and can manage its members and settings.
- **Administrator:** manages user accounts and roles and monitors system health; has technical knowledge.

2.4 Operating environment

Client: modern desktop/mobile browsers (latest two versions of Chrome, Firefox, Edge, Safari). Server: Linux host or container platform running Node.js (Express), with a MySQL database and access to an SMTP service. All traffic is carried over HTTPS.

2.5 Constraints

- Must run in a standard web browser with no plug-ins.
- Passwords and personal data must follow secure-storage practices (OWASP guidelines).
- Open-source technology stack (React + Vite, Node.js/Express, MySQL) to keep cost low.
- Project must be delivered within the academic schedule (see the Test Plan for milestones).

2.6 Assumptions and dependencies

- Users have a stable internet connection and a valid email address.
- An SMTP provider is available for reminder and password-reset emails.
- The server clock is synchronised (NTP) so that due dates and reminders are accurate.

3. External interface requirements

3.1 User interfaces

Responsive web UI with a top navigation bar, project sidebar, list view and Kanban view. Forms provide inline validation messages. Accessibility: keyboard navigation, ARIA labels for screen readers, sufficient colour contrast (WCAG 2.1 AA) and an optional dark / high-contrast theme.

3.2 Hardware interfaces

No special hardware is required. The client runs on any device with a supported browser (PC, laptop, tablet, smartphone). The server requires a standard x86-64/ARM host or cloud VM.

3.3 Software interfaces

- **REST API (JSON over HTTPS)** between the web client and the server for all operations.
- MySQL relational database accessed only by the server through a repository/ORM layer.

- **SMTP service** for reminder and password-reset emails.
- **Logging / monitoring** interface exposing a /health endpoint and structured logs.

3.4 Communications

TLS 1.2+ is enforced for all connections. The API is stateless and uses bearer JWTs. The client retries failed idempotent requests with exponential back-off and shows an offline banner if the server is unreachable.

4. System features (detailed)

Each requirement includes acceptance criteria and a reference test case. IDs follow the pattern TM-F-###. In total this SRS defines 20 functional requirements, 7 non-functional requirements, 4 security objectives and 7 security requirements.

4.1 Authentication

Description: Allow users to create accounts and securely access the system. Passwords are stored only as salted hashes; sessions use short-lived tokens.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-001	The system shall allow a guest to register an account using name, unique email and a password meeting the password policy (min 8 chars, letters + digits).	Functional	High	Product Owner	AC-TM-F-001: Valid data creates an account; a duplicate email is rejected with a clear message. Test: TC-Auth-01	Email format validation
TM-F-002	The system shall authenticate a registered user by email and password and start a session (log in) and allow the user to log out.	Functional	High	Security	AC-TM-F-002: Correct credentials open the dashboard; wrong credentials show a generic error; logout invalidates the token. Test: TC-Auth-02	Depends on TM-F-001
TM-F-003	The system shall temporarily lock an account for 15 minutes after 5 consecutive failed login attempts and record an audit event.	Functional	High	Security	AC-TM-F-003: 5th failure blocks further attempts and shows a lock message; event stored in audit log. Test: TC-Auth-03	Links to TM-SR-006
TM-F-004	The system shall let a user reset a forgotten password through a single-use email link that expires after 30 minutes.	Functional	Medium	Product Owner	AC-TM-F-004: Link resets the password once; expired or reused link is rejected. Test: TC-Auth-04	Requires SMTP service

4.2 Task Management

Description: Core create, read, update and delete operations on tasks together with status and assignment handling.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-010	The system shall allow a user to create a task with a title (required, max 150 chars), description, due date and priority (Low/Medium/High).	Functional	High	Business	AC-TM-F-010: Task is saved and shown in the list; empty title is rejected. Test: TC-Task-01	Authenticated user
TM-F-011	The system shall allow the task owner or assignee to edit any task field and shall record each change.	Functional	High	Business	AC-TM-F-011: Edited values persist and appear in the history. Test: TC-Task-02	Links to TM-F-032
TM-F-012	The system shall let a user delete a task after confirmation; deleted tasks are soft-deleted and recoverable for 30 days.	Functional	Medium	Business	AC-TM-F-012: Task disappears from lists after confirmation; restore within 30 days succeeds. Test: TC-Task-03	Scheduled purge job

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-013	The system shall support task status values To Do, In Progress and Done and shall store a completion timestamp when a task becomes Done.	Functional	High	Business	AC-TM-F-013: Status change is reflected in all views; timestamp set on Done. Test: TC-Task-04	Feeds TM-F-031
TM-F-014	The system shall allow a task to be assigned to any member of the same project and notify the assignee.	Functional	High	Team Lead	AC-TM-F-014: Assignee sees the task under My Tasks and receives a notification. Test: TC-Task-05	Requires TM-F-020

4.3 Projects, Organisation & Reminders

Description: Group tasks into projects, categorise them, find them quickly and remind users of deadlines.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-020	The system shall allow a user to create, rename, archive and add members to projects.	Functional	High	Team Lead	AC-TM-F-020: Project appears for all its members; archived projects become read-only. Test: TC-Proj-01	Project owner role
TM-F-021	The system shall allow tasks to be labelled with one category and multiple tags.	Functional	Medium	Business	AC-TM-F-021: Category and tags are saved and displayed on the task card. Test: TC-Proj-02	-
TM-F-022	The system shall provide keyword search and filtering by status, priority, due date, assignee and tag, with sorting by due date or priority.	Functional	High	Business	AC-TM-F-022: Results match all selected filters and return within 2 s. Test: TC-Srch-01	Indexed columns
TM-F-023	The system shall send in-app and email reminders 24 hours before and on the due date, and flag overdue tasks visually.	Functional	Medium	Business	AC-TM-F-023: Reminders are delivered at the configured times; overdue tasks show a red marker. Test: TC-Rem-01	Scheduler + SMTP

4.4 Progress Tracking

Description: Visual tools to organise and track work progress.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-030	The system shall present tasks on a Kanban board with To Do, In Progress and Done columns and allow drag-and-drop status changes.	Functional	High	Product Owner	AC-TM-F-030: Dropping a card in another column updates its status immediately. Test: TC-Brd-01	Uses TM-F-013
TM-F-031	The system shall show a dashboard with task counts per status, percentage complete per project and the number of overdue tasks.	Functional	High	Product Owner	AC-TM-F-031: Figures match the underlying data at load time. Test: TC-Dash-01	Aggregation queries
TM-F-032	The system shall keep an activity history for every task (who, what changed, when).	Functional	Medium	Audit	AC-TM-F-032: Every create/edit/status/assignment action appears in chronological order. Test: TC-Hist-01	Immutable records

4.5 Collaboration & Administration

Description: Team collaboration, data export and administrative controls.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-040	The system shall allow project members to add comments to a task.	Functional	Medium	Team Lead	AC-TM-F-040: Comment is saved with author and timestamp and visible to all project members. Test: TC-Cmt-01	-
TM-F-041	The system shall allow an administrator to create, deactivate and change roles (Admin/Member) of user accounts.	Functional	High	Operations	AC-TM-F-041: Deactivated users cannot log in; role change takes effect on next request. Test: TC-Adm-01	Admin role only
TM-F-042	The system shall export the currently filtered task list to a CSV file.	Functional	Low	Business	AC-TM-F-042: File opens in a spreadsheet and contains exactly the filtered rows. Test: TC-Exp-01	Uses TM-F-022
TM-F-043	The system shall restrict access to a project and its tasks to the members of that project and to administrators.	Functional	High	Security	AC-TM-F-043: A non-member receives 403 when requesting project data. Test: TC-Adm-02	Server-side check

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test cases. IDs follow the pattern TM-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
TM-NF-001	Page load and UI interaction shall complete within 2 s for 95% of requests and API responses within 500 ms (95th percentile) under 500 concurrent users.	Performance	High	95th percentile within limits in JMeter run. Test: TC-Perf-01
TM-NF-002	The system shall be available 99.5% per month, excluding scheduled maintenance.	Reliability	High	Uptime monitoring report >= 99.5%. Test: Ops report
TM-NF-003	User passwords shall never be stored or logged in plaintext; personal data shall be protected in transit and at rest.	Security	High	Database inspection and log review show no plaintext secrets. Test: TC-Sec-02
TM-NF-004	The UI shall be responsive (desktop, tablet, mobile) and conform to WCAG 2.1 AA where applicable.	Usability / Accessibility	Medium	Accessibility audit (axe/Lighthouse) passes. Test: TC-UX-01
TM-NF-005	The system shall support 1,000 concurrent users and 100,000 tasks without more than 20% degradation of response times.	Scalability	Medium	Load test at 1,000 users meets target. Test: TC-Perf-02
TM-NF-006	The web client shall work on the latest two versions of Chrome, Firefox, Edge and Safari.	Compatibility	Medium	Smoke suite passes on all listed browsers. Test: TC-Comp-01
TM-NF-007	Database backups shall run daily (RPO <= 24 h) and service shall be restorable within 4 hours (RTO).	Recoverability	Medium	Restore drill completed within 4 h. Test: TC-OPS-01

5.1 Security

5.1.1 Security Objectives

- **Confidentiality** - Only authenticated and authorised users can view or modify task and project data; credentials and personal data are protected in transit and at rest.
- **Integrity** - Task data can be changed only through validated, authorised operations and every change is traceable to a user.
- **Availability** - The service resists abuse such as brute-force login and request flooding so legitimate users can keep working.
- **Accountability** - Security-relevant events (login, lock-out, role changes, deletions) are logged with timestamps for audit and investigation.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
TM-SR-001	All client-server communication shall use TLS 1.2 or higher; plain HTTP shall redirect to HTTPS.	Security	High	AC: SSL scan shows no protocol below TLS 1.2. Test: TC-Sec-01
TM-SR-002	Passwords shall be hashed with bcrypt (cost >= 10) with a unique salt per user.	Security	High	AC: DB shows only bcrypt hashes. Test: TC-Sec-02
TM-SR-003	Access tokens (JWT) shall expire after 30 minutes; refresh tokens after 7 days and shall be revocable at logout.	Security	High	AC: Expired or revoked token returns 401. Test: TC-Sec-03
TM-SR-004	Role-based access control shall be enforced on the server for every API call (Admin vs Member; project membership).	Security	High	AC: Unauthorised calls return 403 even if the UI is bypassed. Test: TC-Sec-04
TM-SR-005	All inputs shall be validated and sanitised; database access shall use parameterised queries and output shall be encoded to prevent SQL injection and XSS.	Security	High	AC: OWASP ZAP scan reports no High findings for injection/XSS. Test: TC-Sec-05
TM-SR-006	Login and password-reset endpoints shall be rate-limited to 10 requests per minute per IP.	Security	Medium	AC: 11th request in a minute returns 429. Test: TC-Sec-06
TM-SR-007	Security events (login success/failure, lock-out, role change, deletion) shall be logged with user, time and IP, and retained for 1 year without sensitive data.	Security	Medium	AC: Events present in log; no passwords or tokens present. Test: TC-Sec-07

6. Quality attributes & Acceptance tests

- **Exit criteria for acceptance:** all High-priority functional requirements are implemented and verified, no critical or high-severity defects remain open, all NFR targets in Section 5 are met, and the RTM shows every test case as passed.
- **Acceptance test suites:** Authentication, Task Management, Projects & Search, Progress Tracking, Collaboration & Admin, Performance, Security, Accessibility and Compatibility.
- **Quality attributes emphasised:** usability (fast task entry, clear progress), security (protected accounts and data), reliability (99.5% availability) and maintainability (layered, modular design).
- **Automated testing:** Vitest for JavaScript modules and React components, and Supertest for REST API endpoint tests (Express), run through npm scripts.

7. System models and diagrams

7.1 UML Use-Case diagrams

Diagram 1 shows account and task-management use cases for Guests, Registered Users and Administrators. Diagram 2 shows collaboration and progress-tracking use cases including the external Notification Service actor.

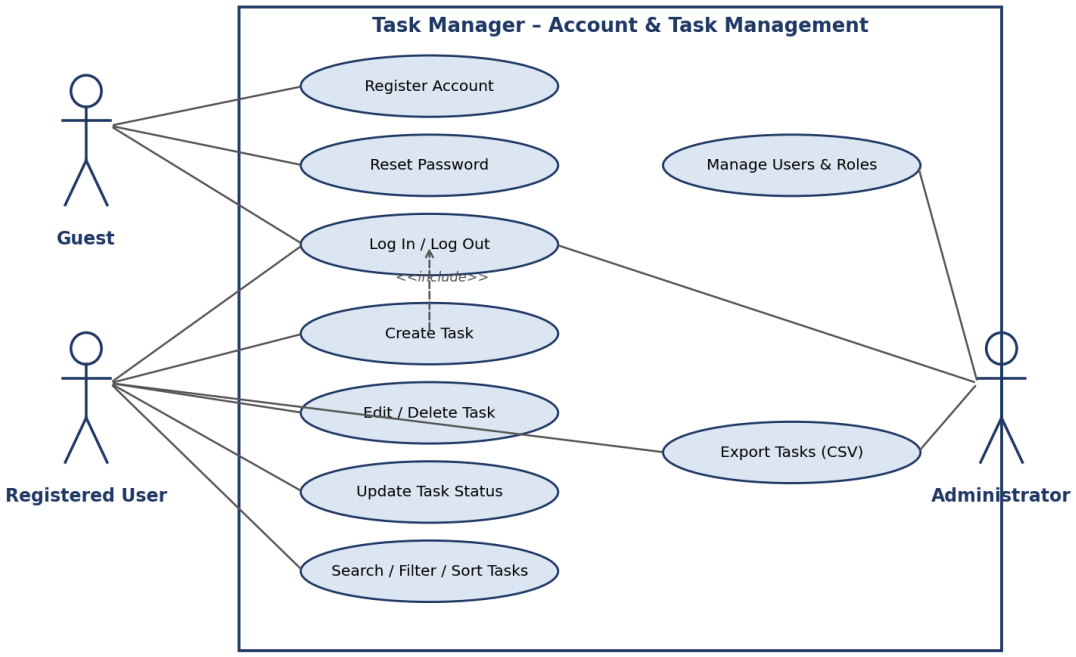


Figure 1 - Use-Case Diagram: Account & Task Management

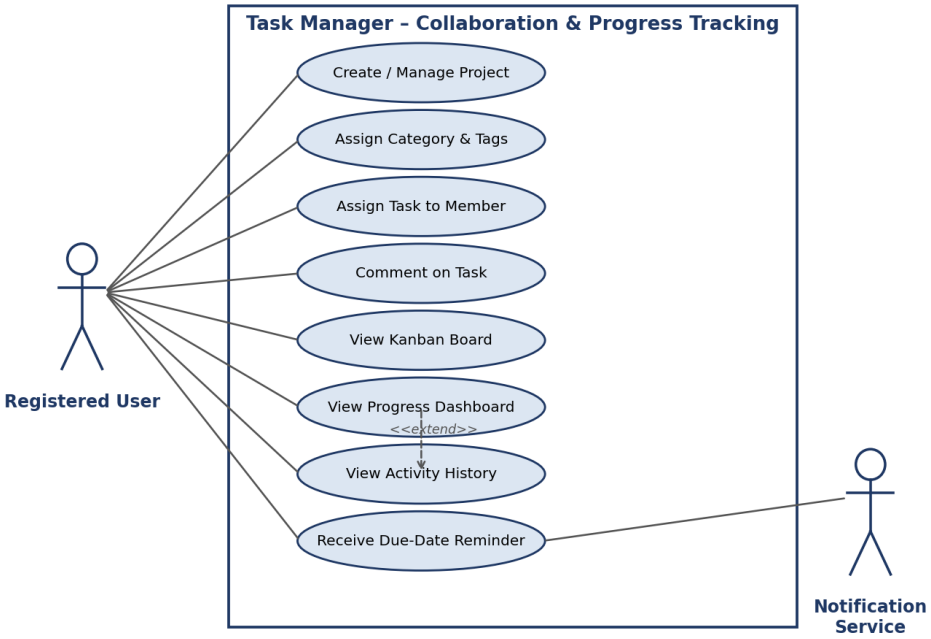


Figure 2 - Use-Case Diagram: Collaboration & Progress Tracking

8. Requirements Traceability Matrix (RTM)

Status legend: N = Not started, P = In progress, A = Accepted / verified. Design Spec references (DS-...) correspond to sections of the Software Architecture and Design document.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
TM-F-001	Register account	4.1 / DS-Auth-01	AuthService	TC-Auth-01	N	
TM-F-002	Login / logout	4.1 / DS-Auth-02	AuthService	TC-Auth-02	N	
TM-F-003	Account lock-out	4.1 / DS-Auth-03	AuthService	TC-Auth-03	N	
TM-F-004	Password reset	4.1 / DS-Auth-04	AuthService, NotificationService	TC-Auth-04	N	
TM-F-010	Create task	4.2 / DS-Task-01	TaskService	TC-Task-01	N	
TM-F-011	Edit task	4.2 / DS-Task-02	TaskService	TC-Task-02	N	
TM-F-012	Delete / restore task	4.2 / DS-Task-03	TaskService	TC-Task-03	N	
TM-F-013	Task status	4.2 / DS-Task-04	TaskService	TC-Task-04	N	
TM-F-014	Assign task	4.2 / DS-Task-05	TaskService, NotificationService	TC-Task-05	N	
TM-F-020	Manage projects	4.3 / DS-Proj-01	ProjectService	TC-Proj-01	N	
TM-F-021	Category & tags	4.3 / DS-Proj-02	ProjectService, TaskService	TC-Proj-02	N	
TM-F-022	Search & filter	4.3 / DS-Srch-01	TaskService	TC-Srch-01	N	
TM-F-023	Reminders	4.3 / DS-Rem-01	NotificationService	TC-Rem-01	N	
TM-F-030	Kanban board	4.4 / DS-Brd-01	Task Board UI, TaskService	TC-Brd-01	N	
TM-F-031	Progress dashboard	4.4 / DS-Dash-01	ReportService, Dashboard UI	TC-Dash-01	N	
TM-F-032	Activity history	4.4 / DS-Hist-01	ReportService, TaskService	TC-Hist-01	N	
TM-F-040	Task comments	4.5 / DS-Cmt-01	TaskService	TC-Cmt-01	N	
TM-F-041	User administration	4.5 / DS-Adm-01	AuthService, User Admin	TC-Adm-01	N	
TM-F-042	CSV export	4.5 / DS-Exp-01	ReportService	TC-Exp-01	N	
TM-F-043	Project access control	4.5 / DS-Adm-02	ProjectService, AuthService	TC-Adm-02	N	
TM-NF-001	Response time	5 / DS-Perf-01	Web UI / API	TC-Perf-01	N	
TM-NF-002	Availability	5 / DS-Rel-01	Deployment	Ops report	N	
TM-NF-003	Data protection	5 / DS-Sec-01	AuthService / DB	TC-Sec-02	N	
TM-NF-004	Accessibility / responsive	5 / DS-UX-01	Web UI	TC-UX-01	N	
TM-NF-005	Scalability	5 / DS-Perf-02	API / DB	TC-Perf-02	N	
TM-NF-006	Browser support	5 / DS-Comp-01	Web UI	TC-Comp-01	N	
TM-NF-007	Backup & recovery	5 / DS-Ops-01	DB / Ops	TC-OPS-01	N	
TM-SR-001	TLS 1.2+	3.4 / DS-Sec-01	Gateway / API	TC-Sec-01	N	
TM-SR-002	Password hashing	4.1 / DS-Sec-02	AuthService	TC-Sec-02	N	
TM-SR-003	Token expiry	4.1 / DS-Sec-03	AuthService	TC-Sec-03	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
TM-SR-004	RBAC enforcement	4.5 / DS-Sec-04	API middleware	TC-Sec-04	N	
TM-SR-005	Input validation	3.4 / DS-Sec-05	API / UI	TC-Sec-05	N	
TM-SR-006	Rate limiting	4.1 / DS-Sec-06	API gateway	TC-Sec-06	N	
TM-SR-007	Security logging	3.4 / DS-Sec-07	AuditLog	TC-Sec-07	N	